all size flues, gig to hold steam pipe rings, to be ground with air motor, one for drilling and reaming holes on superheater header, standard gages to keep all beading tools, prosser expanders and ball joints to a standard. These should be carefully checked, as an odd size beading tool or prosser may do considerable damage to the flue or flue sheet.

In renewing the flues at first we had to take the large superheater flues to the pipe shop, have the ends cut off in a pipe machine, back to boiler shop and weld on safe ends, back to pipe shop to cut right length. To avoid transferring, and to expedite the work, we had the flue cutting machine remodeled, the old machine worked with hand screw feed for cutting flues; we had new machine made with an air cylinder to feed the cutter through flue, also a clamp or steady rest to hold other end of flue while cutting; this operates with air; they are both operated by one foot pedal located about the center of machine, so the operator does not have to move from center of machine in handling the flues. The welding machine was changed slightly, and a new mandril made for welding the large flues. The flue plant is arranged so flues pass through with one handling; they are rolled out of cleaner on short inclined rails to flue cutting machine, passed on to furnace, then to flue welding, safe ends welded on, passed on to next flue cutting machine, and cut to rail length, passed to hydropneumatic testing machine, tested at 250 pounds pressure, then to grinder to have scale and burr removed, then loaded on car or wagon for delivery to engine or shipment.

In removing flues from boiler, the front end should be cut off as close inside the sheet as possible; the back end should be cut far enough to free it from prosser marks, or if safe ends have been applied they should be cut off to remove the old weld, maintaining only one weld on the swaged end of the flues. In cleaning the flues, care should be taken that there are no rivet heads or projections inside the rattler; the large flues are considerably heavier than the 2-inch or 2¼-inch flue, therefore must be handled more carefully, and any projections will dent the flues or cause the ends to start to crack, making it necessary to cut considerably off the flue in order to square up the end, or if dented and the dents not removed, it will make it difficult to insert the units in the flues.

The safe ending of the large flues is to be done at the firebox end, rather than smoke-box end; this provides new material where service conditions are most severe. It is recommended that the safe end be annealed after applied, in order to prevent liability of cracking and permit it being worked more easily in the flue sheet. When safe ends have been applied so that all space has been utilized between flue sheet and return bends, the flue should be reswaged, and a long safe end or extension applied to the front end of the flue, being careful to leave it smooth with no obstruction on the inside of flue that would prevent or make difficult to insert the units into the flue.

In welding 4½-inch tubing it is a good practice to scarf the safe end for a distance of about 5 to 8 inches and heat the end of the flue; bell-mouth it and insert the safe end while the flue is hot. This practice insures the piece sticking in the flues in proper place until it is heated and in position in the welding machine. It may be found, in attempting to weld the large flues, that trouble will be experienced in bringing the material down at the point of the weld uniformly to the thickness of the flue. The scarfing of the flue tends to lessen the difficulty experienced in bringing the metal down. It is important that the welded portion of the flue be smooth on the inside, thereby removing all obstructions and facilitating the cleaning of the flues.

On account of the weight of the flues, there may be some difficulty experienced in handling them. A heavy flue rest should be provided at the back of the welding machine and furnace, equipped with a spool or thimble, the same to be subjected to adjustment so that the flue may be uniformly supported and kept in line with the mandril and the furnace and the welding machine during the time that the heat is being made and the welding operation carried on. The difficulty in safe ending large flues may thus be readily overcome by the proper methods of handling, heating and a machine designed to do the work satisfactorily.

Among the general difficulties experienced in tube welding, in the past, has been the use of the improper dies, particularly if used on hammer welding machine; on roller machines, roller not properly adjusted, thus preventing the necessary pressure on the material, which must be obtained to secure a good smooth weld. Another trouble has been furnaces that do not heat fast enough, the result being an excess amount of oxidation or loss of material; furnaces where the heat is too harsh and the action of the air blast too severe on the material, resulting in a loss of material before the flue is ready to be taken from the furnace. Good flue welding demands that the flues and safe ends be heated in a furnace that gives a soft non-oxidizing heat, that will heat the material uniformly to a good welding heat as quickly as possible.

INSTALLATION OF SUPERHEATER

In placing the header in the front end, it should be so set that the face will parallel to the center lines of the top row of flues, and each end of the face the same distance above center line of outside flues of the top row. The header supports should be securely bolted to smoke arch to hold header in proper place, and made so that the header will bear upon them throughout its entire width.

Care should be taken that the flue sheet is laid off the proper distance from the center of dry pipe flue sheet hole, and equal distances across; that it corresponds with header, and that units rest near the top of flue when they are tightened in place, that they may have free circulation, and flue cleaning pipe will go under the units. If the joints on pipe do not pull up true to correspond with holes in header, they should be bent or sprung in place and not be allowed to drag in place by turning up on the nuts. When the units are in place and tightened, they should be in the upper part of the flue.

When everything is tightened in place, apply a water test with 250 pounds pressure. After boiler has been steamed up and superheater tested with steam, the bolts should be gone over and tightened finally. Suitable wrenches should be provided that will reach all unit bolts. These wrenches consist of a single socket wrench made to fit nuts and clean unit pipes, with an extension long enough to allow a bar to be used in turning it without interfering with the units and short enough that it may be used without removing the table plate or damper. A peephole should be provided in the side of the smoke arch to permit the inspection of the front end flue sheet, superheater units and ball joint connections without removing the baffle plates.

All superheater flues should be beaded in front flue sheet. The baffle plates should be made to fit tight and should be so constructed that they can be removed without removing the door ring. The damper, when connected to steam chests, should be put up so that when weight is down the damper will be closed; when the weight is raised the damper should be open. When connected to the blower, its operating is reversed. Care should be taken that it operates freely.

DISASTROUS EXPLOSION OF A WATERTUBE BOILER.—Seven men were killed and three seriously injured by the explosion of a boiler at the plant of the Richmond Light & Railway Company, Livingston, Staten Island, N. Y., on October 21. The boiler was of the vertical Wickes type, rated at 300 horsepower. The drums were about 7 feet in diameter and the boiler was carrying a pressure of 150 pounds per square inch. The failure occurred in the bottom head, which fractured on a line close to the flange of the head. The exact cause for the failure of the head has not been determined, as the metal showed no signs of wasting away.

shown greater economy, besides occupying less space, having less weight and with a lower center of gravity than boilers with a larger ratio of heating surface to grate area. Furthermore, the boiler of small ratio requires less material and labor and, consequently, less first cost for its construction and smaller maintenance costs for its upkeep.

Best Methods of Welding Superheating Tubes and Tools Used for Same*

All safe ending should be done at the firebox end, on account of the necessity of always having good and new material at the location where it is needed the most. This is the location of the extreme heat, as well as the stresses that have to be taken care of on account of the great amount of expansion and contraction, as well as the amount of work that has to be done on these flues, at this particular location, to keep the same tight and in good working condition. It is the experience of the roads that have operated these superheating engines the longest that it is absolutely necessary to keep the flues tight at all times to run them successfully. It is also the experience, in bad water districts, that these large tubes give considerable trouble; this necessitates a great deal of work being done on them. Therefore the committee would recommend that the safe end be applied on the firebox end of tubes at all times.

To cut the burrs off the end of the flues and use the body of the flue in the firebox end for a bead, in our opinion, would not give satisfactory results. We all know the body flue deteriorates rapidly in bad water districts when in these boilers and in hard service. Very few, if any, of our roads would think of applying our ordinary 2-inch or 2¼-inch flues in this manner. This being the case, there is no good reason why we should apply our superheating flues with any less care and with material which has been in the boiler as a body flue for some time, under the above-mentioned conditions, which is not as good as a new safe end. The duties of a superheating flue in the firebox are as hard, if not harder, than any ordinary flue, and under some conditions, such as bad water and failure to keep same absolutely clean, they are very liable to give trouble and require considerable work to keep them tight. This being the case they should be safe ended in the firebox end, at all times, with the very best material obtainable.

It is the opinion of some of the committee that there should not be more than one weld in these flues at any time, on account of old welds giving trouble after flues have been in service, removed from boiler, cleaned and replaced again. If the welds are reduced to one in each flue you reduce the liability of having more than one leak from this source in each flue. This also adds to the safety of those who come in contact with and make repairs to this class of work.

To apply the safe end at the firebox end, at all times, it will be necessary to cut your flue as close to the front sheet as possible to avoid any additional waste of the body flue, and reduce the necessity of applying long safe ends. The average life of a body flue of this dimension, in this particular service, would not exceed four safe endings. The average flue welding machines are built so they can weld a safe end 14½ inches long. This can be done by first welding on a 5-inch piece, second an 8-inch piece, third an 11-inch piece and fourth a 14-inch piece. At the expiration of this time unquestionably the body flue will be about worn out. To do this it will be necessary to cut your flue close to the front flue sheet, as well as close to the old weld in the back end. In repairing your flue in this manner you will always have a new safe end at the firebox end, as well as a flue with only one weld at any time.

This will give an almost perfect flue in the locality where it is required to do the greater part of the work; also, where it needs all the strength and durability it is possible to have at all times. In some localities these flues are prepared and welded in the following manner:

The rough end of the flue at the firebox is cut off on the cutting-off machine by beveled cutters, which leaves the end of the body flue beveled to an angle of 32 degrees. The flue

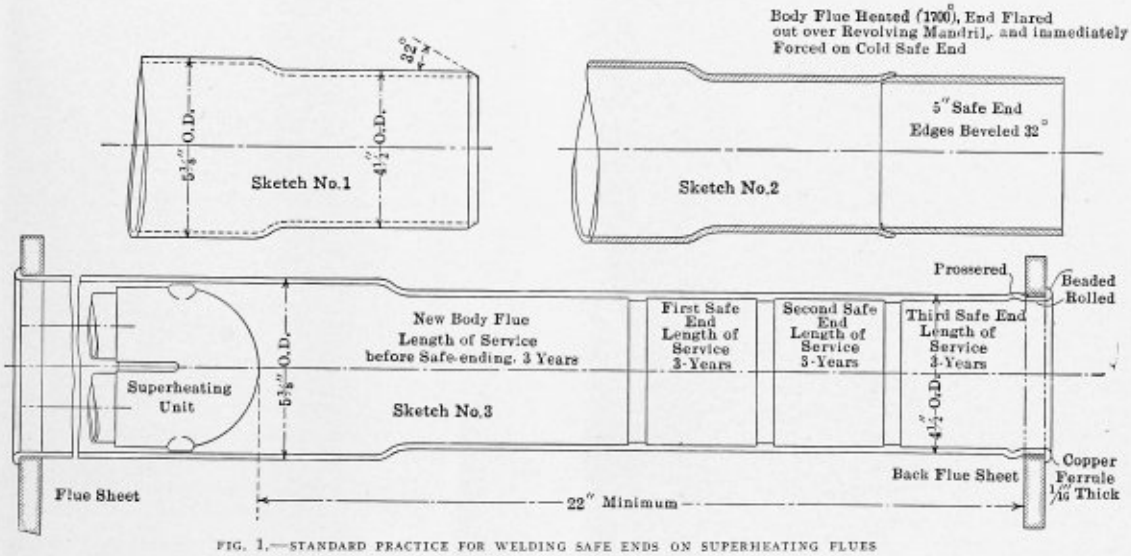


FIG. 1.—STANDARD PRACTICE FOR WELDING SAFE ENDS ON SUPERHEATING FLUES

* Abstract of a committee report presented before the Master Boiler Makers' Association at Chicago, May, 1913, by B. F. Sarver, J. J. Orr and J. P. Malley.

is then heated and flared out on a revolving mandril sufficiently to admit of the insertion of a (cold) standard safe end. The safe ends are cut off 5 inches long with the same type cutters as previously noted, thereby giving a sufficient angle for scarfing (32 degrees) without any additional preparation.

The 5-inch safe ends are placed on a mandril, and, after the body flue is heated and flared out, it is forced by hand over this (cold) safe end from ⅜-inch to ½-inch. This process gives a good scarfing edge, as both pieces have been beveled to the same angle, 32 degrees, and when the body flue "cools" it obtains a positive grip on the safe end, due to its contraction; this also positions the safe end, so that a good welding heat is obtained.

The flue with safe end attached is then heated in a furnace, especially prepared so as to give a narrow flame heat on the joint to be welded. The flue is heated to a welding heat, but

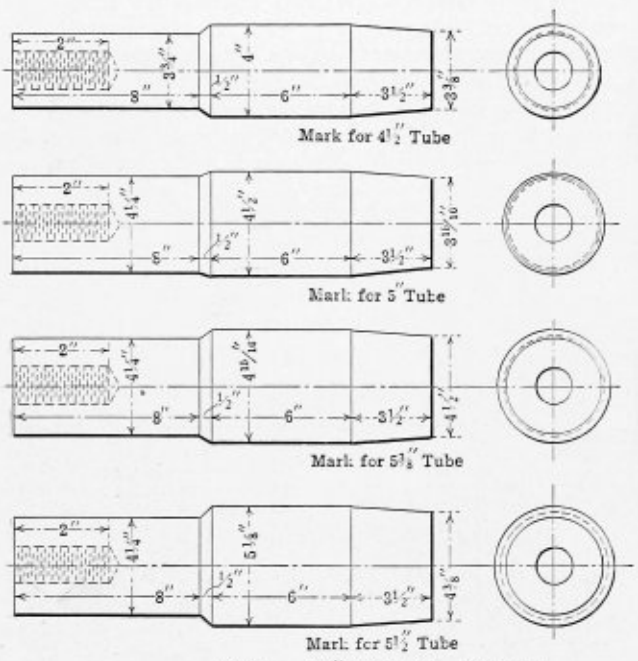


FIG. 2.—MANDRILS

before being removed from the furnace, and when the heat is proper for welding, they are forcibly bumped up against a water-cooled iron block, inserted in the rear wall of the furnace, or, more technically speaking, the block is located in the rear hearth opening of the furnace.

This bumping action practically welds the flue while in the fire, and the narrow flame eliminates the danger of "heating up" previous welds. The flue is now removed from the furnace and welded on a welding machine, which revolves around the flue, and is specially equipped for this purpose. This completes the operation of applying the safe end.

Since the firebox ends of these flues are swedged down from 5⅜ inches to 4½ inches it is most economical, from any point of view, to apply the safe ends to the firebox end of the body flue, and at the same time it renews this end of the flue, which is subject to the greatest strains from cold air and heat. In some localities it is permissible to have more than one weld in these superheating flues, as shown in the illustrations.

The best tools for welding these flues is a question that is in about the same position at the present time as the welding of the flue. In a great many localities the dimensions of the old welding machines have been increased regardless of the type, and it is claimed by all those who are concerned that they are getting satisfactory results.

At a great many places a welding machine is used that revolves around the flue, and, from reports, successfully; but, like others, some fault is found with them. Fig. 2 shows the proper measurements of different size mandrils to be used in this type of welding machine for different size flues.

Several questions are debatable relative to these machines. They are as follows: Whether the flue should revolve around on the roller in the machine, or whether the machine should revolve around the flue. To revolve the flue around on the roller it requires some time for the flue to travel this distance, and being of light material it is very likely to become cooled off some before the entire weld is made. It also requires a great deal of power and labor to operate and weld these flues successfully with a machine of this type.

With a machine that revolves around the flue the weld is made a great deal quicker and with less power and labor, but this machine also has its faults. First, it is claimed that it does not make a smooth job on the inside, but leaves an obstruction in the flue that is likely to cause the flue to clog rapidly. This is a very serious objection, as it is the opinion of almost all that it is necessary to have these flues clean at all times. It is also claimed that it is hard to weld the safe end on these large flues and keep them straight with this machine.

After taking into consideration the above-mentioned conditions your committee does not feel at this time that it would be doing justice to itself or any of the machines used for doing this work at the present time to make a complete report on this subject.

In summing up this question, there are three or four points which should receive careful consideration.

First—At which end of body flue should the safe end be applied?

Second—How many welds in a flue are permissible at any one time?

Third—Should welding machines revolve around the flue, or should the flue revolve around the machine?

Fourth—Flues should be annealed at the firebox end at all times.

New Enterprise in the Boiler Making Field

No doubt our subscribers will be interested to learn of a new enterprise in the boiler making field. The latest is the Downingtown Iron Works, Inc., of Downingtown, Pa. The three young men back of this enterprise are all practical boiler makers and know the business from A to Z, having had a wide experience. Mr. Louis C. Zimmerman is president and general manager, Mr. Harold T. Abele is vice-president and Mr. Park L. Plank is secretary and treasurer. The works are splendidly located on the main line of the Pennsylvania Railroad within easy reach of Philadelphia, and a short distance from the steel plate mills at Coatesville, Pa., where supplies necessary for all kinds of boiler work can be had at short notice. The present shop is 90 feet long, 40 feet wide and 18 feet high, constructed of brick throughout. The machines are all operated by electricity. Overhead cranes are provided for handling all material, finished or unfinished, and all work is done off the ground floor. The working capacity is exceedingly well provided for with plenty of extra space to make additions to the building when necessity requires same. As we are personally acquainted with these gentlemen and know that they are endowed with push and energy, we feel sure that this enterprise will be a complete success.